

CS/SSW 555 Agile Methods

Cyber Guru Crew (Team 17)

Sprint 4

4/23/2024

Project Background Information

Selected Project Option Number: Option 3

Main Objective of Project:

Develop a mobile application that uses AI to detect lies through voice stress, facial expressions, and response timing analysis. This application is aimed at professional and personal use, as well as it promising to enhance truth verification processes with user-friendly features.

Team Members and Roles:

Name	Role
Octavio Morales	Software Developer
Brandon Lopez (did not contribute to this sprint)	Machine Learning Analyst
Bhagawat Chapagain	Software Developer
Edmund Yuen (did not contribute to this sprint)	Machine Learning Analyst
Yunfei Tang (did not contribute to this sprint)	UX/UI Designer

List & Description of Each Sprint:

Sprint #	Objective
Sprint 1	Project onboarding and setup: - Define database schema and set up the database - Choose and set up testing framework - Basic data collection for initial model training
Sprint 2	UI/UX development - Design user interface for the app - Implement user interactions and flow
Sprint 3	Machine Learning Model development: - Select appropriate machine learning model - Train the model with collected data - Evaluate the performance of the model
Sprint 4	Machine Learning Model development:

	• Fine-tune the machine learning model • Integrate the UI components with the machine learning model
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Project Links:

Type	URL
GitHub/GitLab/Etc	https://github.com/octavio-morales1/CS555Team17.git
Jira	https://stevens-team17.atlassian.net/jira/software/projects/KAN/boards/1#
Live Demo (if applicable)	N/A

Technology Stack Used:

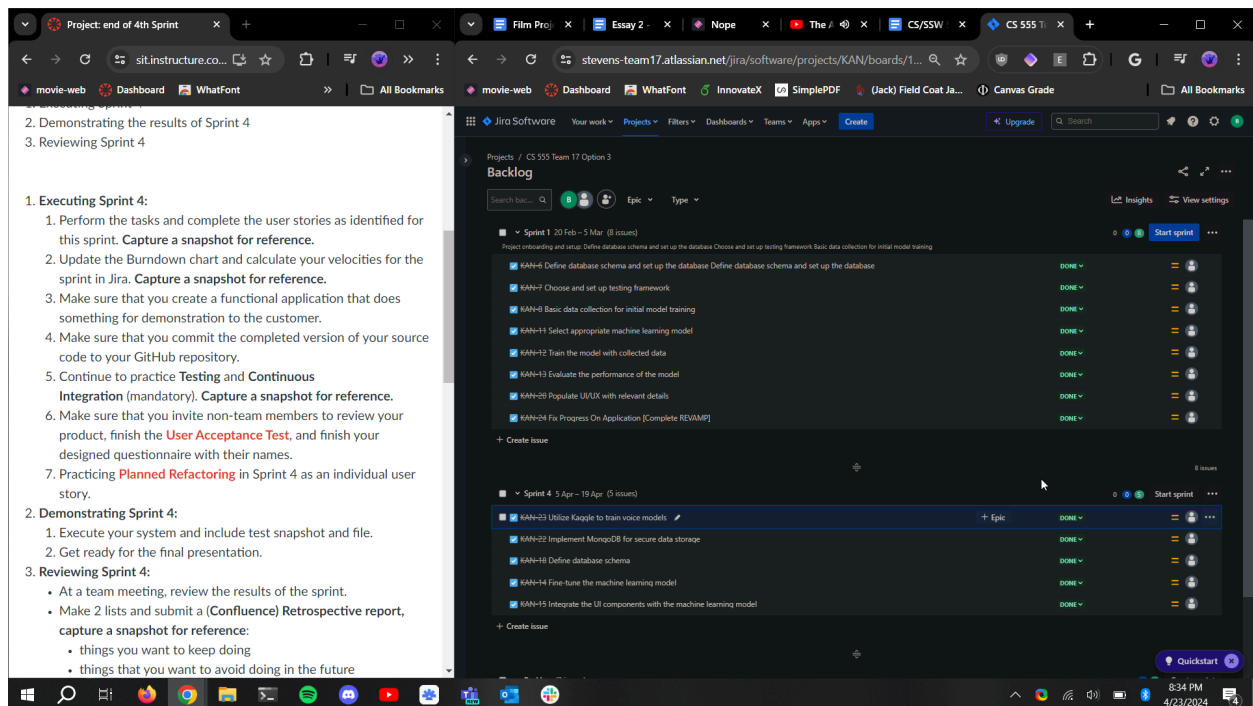
Type	Technology
Data Preprocessing	MatLab
UI/UX	PyQt5 (Python GUI)
Machine Learning	Kaggle
Database	MongoDB

Sprint Progress Update: Current Status

Summary of Work Completed

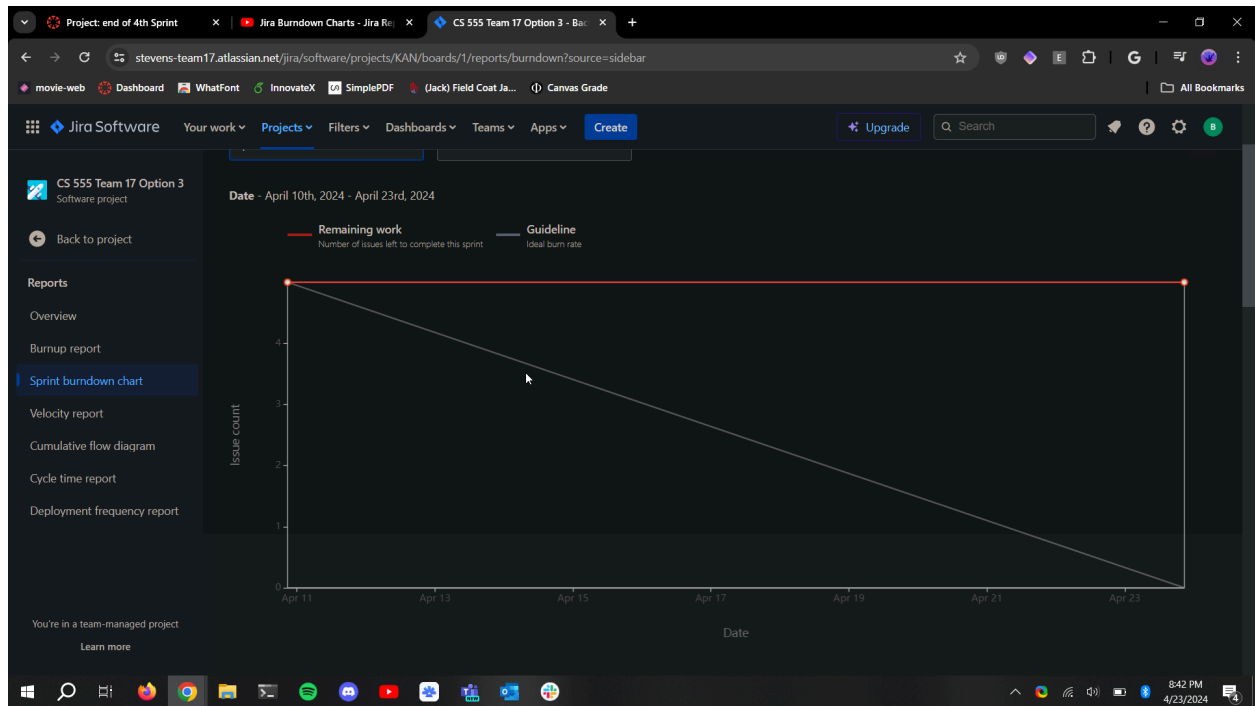
It's clear that our group is behind compared to others in our project. We thought we were making good progress on the lie detector, but the professor sees us as just starting. This means we need to quickly reassess our plan and work more efficiently. We'll have to look at how we're doing things, figure out where we can improve, and set clear goals. It's also important to communicate better within our team and maybe get advice from the professor or other groups (when permitted). This way, we can catch up and meet the project's expectations more effectively. Luckily, our group has believed that we have achieved such progress, putting us at par with the other groups.

Screenshot of Jira Backlog



Screenshot of Burndown Chart

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< > Denotes Example. Must Replace.

UAT:

Here are the responses from the five users after using our lie detector application:

Neel

"I'm impressed with how quickly it processed the audio files. But, I found that the application sometimes gives inconsistent results with the same audio sample. Is there a way to make the predictions more stable?"

Sam

"I like the simplicity of the UI. It would be great if there was a feature to view a detailed report of the analysis, like what specific features led to a 'lie' or 'truth' prediction."

Adelina

"I had a bit of trouble understanding how to get started initially. Maybe adding a brief tutorial or help section on the main screen would make it more user-friendly for first-time users."

Disha

"The application works smoothly, but I encountered an error when I tried to upload a very large file. Perhaps there should be some information about file size limitations or optimization for larger files."

Divya

"It's a fascinating application! I tested it during a group game night, and it was quite a hit. One suggestion: could there be an option to record audio directly through the app? That would enhance the experience."

Planned Refactoring:

- Refactor ML model to be able to handle larger file sizes
- Refactor UI to make it easier for users to be able use the program
- Refactor UI to include detailed analysis including an option to export to PDF
- Refactor UI to include ease-of-use features (even if for now they don't function) like an option to record audio in the app

Confluence Items:

The image displays two screenshots of the Atlassian Confluence interface, showing a draft page titled "End of Sprint 4 - things we want to stop doing" and another titled "End of Sprint 4 - things we want to keep doing".

Top Screenshot: The page title is "End of Sprint 4 - things we want to stop doing". The content lists two items:

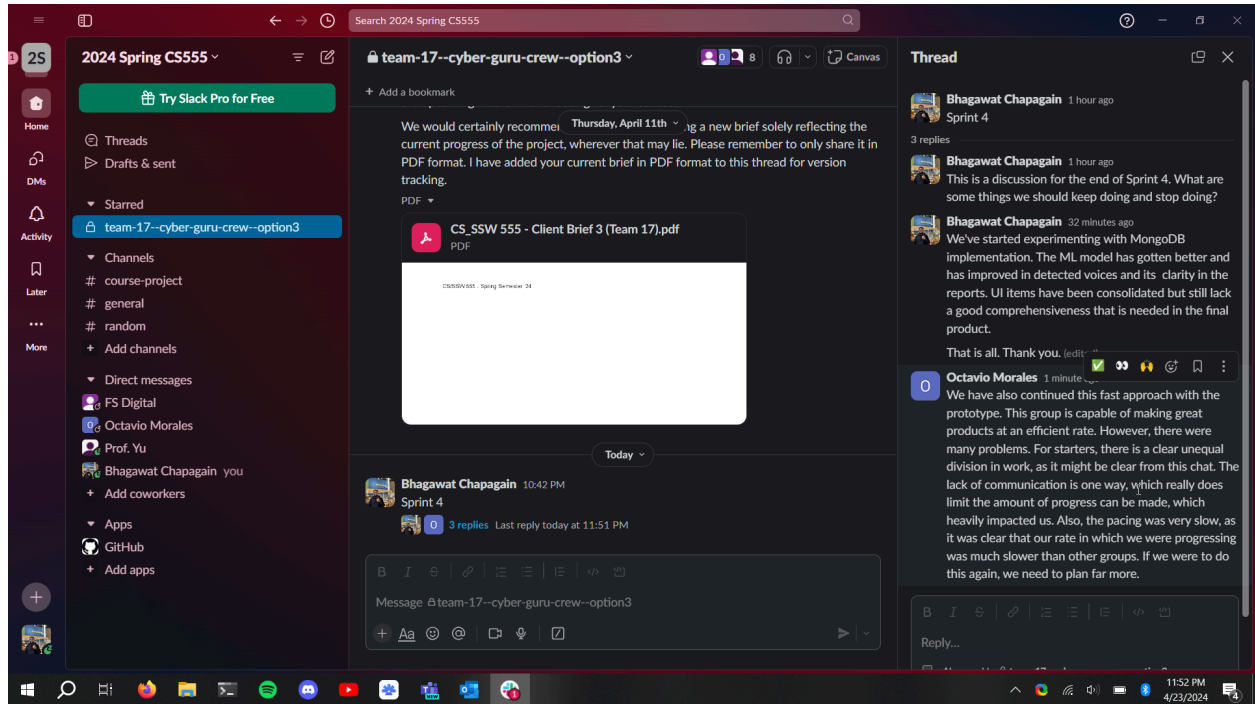
- Low communication – a lot of the team members are still not communicating in the slack which is slowing production down significantly
- UI/UX - get rid of unnecessary UI items

Bottom Screenshot: The page title is "End of Sprint 4 - things we want to keep doing". The content lists three items:

- Working on the effectiveness of the ML model
- Improving on UI/UX design so that customers/stakeholders have an easier time learning how to use the software and the utility it serves
- Improve database storage and ML parsing so that larger file sizes are better supported. Users have reported that their recordings can get very big so we need something to avoid this

Sprint 4 Meeting:

< > Denotes Example. Must Replace.



Video of Project at Current State:

It does not need to be one edited video. If there are multiple videos, please link all of them below:

Public Video URL (Google Drive, Loom, Etc)
No video present as this is not the current stage for visual/audio results.

Sprint Progress Update: Current Blockers

Sprint Progress Update: Current Blockers

Despite our efforts, we've encountered a few hurdles that have slowed our progress during this sprint:

ML Model Scalability: One of the primary blockers we've faced is the scalability of our machine learning model. Handling larger file sizes has proven to be a challenge, leading to errors and inconsistencies in predictions.

UI Complexity: While users appreciate the simplicity of our UI, there's still room for improvement. We've identified complexities in the UI flow that hinder user experience, particularly in understanding how to initiate and interpret the analysis.

Error Handling for File Uploads: Users have reported encountering errors when attempting to upload larger files. This indicates a need for better error handling and perhaps file size limitations to prevent crashes and ensure smoother user interactions.

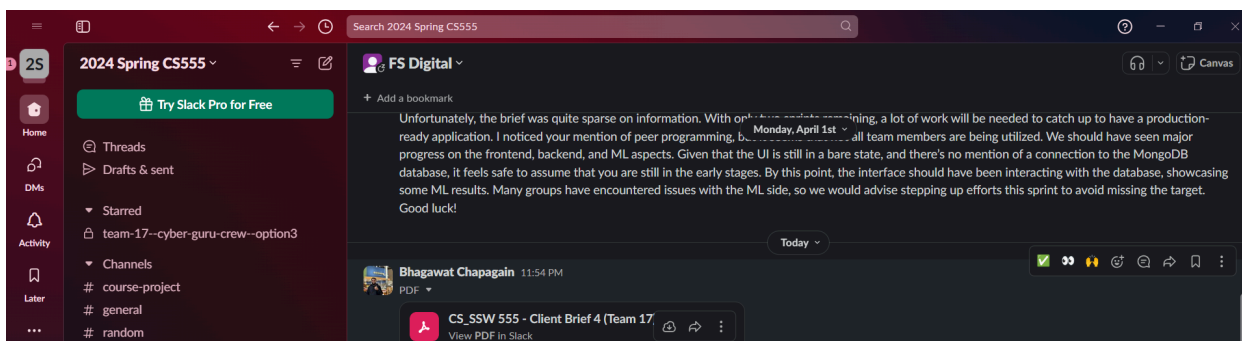
Looking Ahead: Next Sprint

In the upcoming sprint, we plan to address these blockers head-on and make significant strides towards our project goals:

ML Model Refactoring: Our priority will be to refactor the machine learning model to improve scalability and stability. This involves optimizing the model to handle larger file sizes and enhancing prediction consistency.

UI Enhancement: We'll focus on refining the user interface to streamline the user experience. This includes simplifying the UI flow, implementing clearer instructions or tutorials for first-time users, and adding features for detailed analysis feedback.

Error Handling and Optimization: To mitigate issues with file uploads and improve overall performance, we'll implement robust error handling mechanisms and consider optimizations for handling large files more efficiently.



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